MIL-P-82534(OS)
Amendment -2
2 May 1969
Superseding
Amendment-1
26 November 1968

MILITARY SPECIFICATION

PROGRAMMER, RANGE AND AIRFRAME SEPARATION

MARK 3 MOD 1

This amendment forms a part of Military Specification MIL-P-82534(OS) of 18 April 1968 and has been approved by the Naval Ordnance Systems Command, Department of the Navy

Wherever it appears in specification delete "Bureau of Naval Weapons" and substitute "Naval Ordnance Systems Command"; also delete "BuWeps" and substitute "NAVORD".

Page 10, Table III. Make the following changes:

From:	Humidity	4.5.1.2
	Quality Conformance Inspection	4.6
To:	Humidity	4.5.1.2
	Quality Conformance Inspection	4.6**
From:	Vibration, Standby Condition	4.5.1.4
	Quality Conformance Inspection	4.6
To:	Vibration, Standby Condition	4.5.1.4
	Quality Conformance Inspection	4.6**
From:	High Extreme	4.5.1.6.2
	Quality Conformance Inspection	4.6
	Operation	4.6.1.4
	Quality Conformance Inspection	4.6*
To:	High Extreme	4.5.1.6.2
	Quality Conformance Inspection	4.6
	Operation	4.6.1.4
	Quality Conformance Inspection	4.6* **

Add the following note below table:
**All tests of paragraph 4.6 shall be conducted except the enclosure test (4.6.1.5).

FSC 1340

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Page 12, paragraph 4.4.2(c), under "Capability". Delete and substitute "Automatically control displacement/acceleration amplitude of fundamental vibration input to RASP at 0.10-inch double amplitude or 10g, whichever is less, over the frequency range of 20 to 2000 cps. Monitor cross-axis motion of RASP in two directions."

Pages 13 and 15, paragraph 4.5.1.5. Delete and substitute:

"4.5.1.5 Vibration, operating condition. The programmer shall be subjected to sinusoidal vibration over the frequency range of 20 plus or minus 3, to 2000 plus or minus 40 cps, and at a double displacement amplitude of 0.100 plus or minus 0.015 inch or a zero-to-peak acceleration of 10 plus or minus 1.5g, whichever is less, in each of the three axes shown in Figure 1. The frequency range shall be traversed at a logarithmic rate in not less than 2.5 minutes, once in each axis. The vibration system used shall meet the performance requirements specified in 4.5.1.5.1 through 4.5.1.5.3. Perform tests a, b, and c, as specified in 4.6.1.4.4 and 4.6.1.4.5 at room ambient temperature only, not less than twice in each axis. Readings shall be taken at approximately equal time intervals over the entire frequency range. The readback time tolerance of each individual reading shall be plus or minus 0.05 second for channel one and plus or minus 0.80 second for channel two.

4.5.1.5.1 Programmer support requirements. For tests specified in 4.5.1.5, the programmer shall be supported only in the area shown in Figure 2 except that the free ends of the programmer payload cable and test cable may be supported.

4.5.1.5.2 Vibration system requirements. The vibration system used to conduct tests of 4.5.1.5 shall have a continuous duty force rating of at least 125 percent of the calculated force rating required to vibrate the programmer and supporting fixtures at specified levels. Total harmonic distortion in the acceleration waveform over the test range shall not exceed 10 percent. The acceleration system noise level at the unloaded vibration machine exciter shall be less than 0.4g rms with exciter power supply and field energized, and oscillator gain control set for maximum attenuation. The displacement system noise level under the foregoing conditions shall not exceed 0.005-inch double amplitude.

4.5.1.5.3 Vibration control and monitor requirements

4.5.1.5.3.1 Vibration machine output. Output of vibration machine shall be controlled by an automatic amplitude servo system and shall be maintained at the levels of displacement or acceleration specified in 4.5.1.5 by controlling the amplitude of the fundamental frequency.

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Any distortion of the fundamental signal, regardless of source, shall not be allowed to affect the measurement of displacement or acceleration amplitude of the fundamental input frequency to the vibration machine. Such distortion shall be eliminated from the input control signal by use of a tracking band pass filter between the input control circuit and the amplitude servo system.

4.5.1.5.3.2 Vibration pickups. During tests specified in 4.5.1.5 the output of the vibration machine shall be controlled and monitored by three piezoelectric accelerometers mounted in the vicinity of the programmer so as to indicate the greatest motion of the supporting fixture on which the programmer is directly mounted. The accelerometers shall be positioned in three mutually orthogonal axes. One accelerometer, mounted with its axis parallel to the axis of motion, shall be used to control power to the vibration machine. The remaining two accelerometers, mounted parallel and perpendicular to the vibration machine trunnion axis, shall be used to monitor cross-axis motion. Unfiltered cross-axis motion as indicated by either accelerometer shall not exceed the filtered input to the vibration machine. Motion transducers other than piezoelectric accelerometers shall not be used for input control or monitoring.

4.5.1.5.3.3 Control and monitor circuits. Circuits used to control and monitor the vibration system for tests specified in 4.5.1.5 shall meet the following requirements:

(a) Displacement circuits shall indicate amplitude of input displacement to the programmer over the range of 0.010-inch double amplitude to the upper limit of the equipment to an accuracy of plus or minus 10 percent from 10 to 100 cps.

(b) Acceleration circuits shall indicate amplitude of input acceleration to the programmer to an accuracy of plus or minus 10 percent over the range of 17 to 2040 cps. The circuits shall be capable of indicating the amplitude of both the fundamental and unfiltered signal. It shall not be necessary, however, to indicate these signals simultaneously."

Page 15, paragraph 4.6.1.1, last sentence: Delete "an indicated" and substitute "as indicated".

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